Research Article

A Residual Deep Learning Framework for EEG-Based Epileptic Seizure Stage Classification

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Abstract: Epilepsy is a problem that affects more than 70 million people all around the world. It puts a lot of pressure on healthcare systems. For doctors who take care of epilepsy patients it is very important to know where the patient is in the seizure cycle. Before, during or between seizures. The time before a seizure starts is the only time when doctors can still prevent the seizure. Most machines that help doctors only look for seizures after they have started not before they happen. In our study we made a computer program that can look at brain signals and tell if a patient is okay about to have a seizure or having a seizure right now. We called this program a one- residual convolutional neural network. We tried it out on brain signals from ten epilepsy patients at the Neurology & Sleep Centre in Hauz Khas, New Delhi. We recorded the brain signals quickly 200 times per second and made sure they were clear by filtering out any extra noise. Then we broke the signals into overlapping 1.28-second windows to see if our program could tell what was happening. The main focus of this work was to make sure our methods were really good. We split the data into files instead of smaller pieces, which stopped the information from getting mixed up between the groups we used to train and test our model. This is a problem in this field. We only used the training data to get the scale for our measurements. Our model was really accurate it got 90.77 percent of the answers overall and it did well on other tests too like the macro F1 score which was 0.907 and the mean AUC-ROC which was 0.982 on the test set that was not used during training. The proposed model achieved near-perfect ictal detection, with an F1 score of 0.996 and an AUC of 1.00. Preictal recall of 0.78 reflects the inherent difficulty of distinguishing preictal EEG from the interictal baseline.

Keywords: Epileptic seizure detection, EEG classification, Residual CNN, Seizure stage detection, Deep learning

1. Introduction

Epilepsy is a condition that affects the brain. It causes seizures and a lot of uncertainty for patients and their caregivers [1, 2]. They do not know when the next seizure will happen. Doctors need to know where a patient is in their seizure cycle to make treatment choices. They can then decide on medication changes, surgery or using a device to help control seizures.

The EEG is the tool for monitoring epilepsy. It is a test that measures the brains activity through electrodes on the scalp. Trained doctors can look at the EEG signals. See patterns that indicate seizures are happening. The problem is that long-term monitoring produces a lot of data and manually reviewing hours of EEG recordings requires a lot of expertise and time that busy clinical settings often cannot spare. Automated classification systems can help with this. Not by replacing doctors. By flagging and categorizing signal segments for clinical review [3].

Seizures go through four phases: the phase before the seizure the seizure itself the recovery phase and the baseline phase [4]. The phase just before the seizure is the important and the most difficult to detect. It is the phase where a timely intervention. Medication or neurostimulation. Can potentially prevent the seizure from fully developing. Despite these automated systems only look at whether the patient is currently seizing or not. This is useful for detection. Not useful enough for clinical management.

Some machine learning approaches have been used to classify EEG seizures. They rely on handcrafted features that are time-consuming to create and do not work well across different patients and seizure types [5–7]. Deep learning is different. It learns features directly from the signal, and various deep learning models have shown strong performance on EEG tasks [8–10, 21, 23, 24]. For instance, specialized architectures like EEGNet [22] and other deep convolutional networks have advanced multi-class seizure classification [25]. Recently, the field has seen the introduction of transformer-based models [26,31], temporal convolutional networks [30], self-supervised learning [35], and explainable artificial intelligence (XAI) frameworks [28, 34]. Furthermore, residual neural networks and multi-scale CNNs have been actively developed for robust and real-time seizure detection [27, 29, 32, 33].

However, there is a problem with the published literature on this topic. The way that training and test data are split is not correct which means that the reported ac-

curacy is not reliable [4, 14]. This paper proposes a new approach to classifying EEG seizures using a type of deep learning model called a 1D-ResNet.

The main contributions of this work are: A 1D-ResNet architecture that works directly on raw EEG time-domain signals without any feature engineering. A protocol for splitting training and test data that eliminates the problem of data leakage. Test results that show accuracy and are competitive with other published approaches while being fully reproducible and reliable.

The results of this study are promising. The model achieved an accuracy of 90.77% and a macro F1 of 0.907 which is very good. The model also detected seizures with perfect accuracy, which is very important for clinical management. Overall, this study shows that deep learning can be used to reliably classify EEG seizures, which could be very helpful for patients with epilepsy and their caregivers.

2. Methods

2.1. Dataset

The dataset we are using for this project is made up of single-channel EEG recordings from ten patients. These recordings were collected at the Neurology & Sleep Centre, Hauz Khas, New Delhi [12, 13, 15–17]. The EEG signals were recorded at 200 Hz using a machine called a Grass Telefactor Comet AS40 amplifier. The electrodes used to record the signals were placed on the scalp according to the 10-20 standard.

The raw signals were filtered to remove noise and artefacts from the EEG signals. The EEG signals were then divided into three categories: interictal, preictal, and ictal. Each category contains fifty MAT-format files. Each file contains 1,024 samples, corresponding to approximately 5.12 seconds of single-channel EEG. The files were further segmented into overlapping epochs of 256 samples (1.28 seconds), using a stride of 64 samples between consecutive windows. The three categories are equally represented, ensuring a balanced class distribution.

2.2. Preprocessing and Data Partitioning

The EEG signals were provided in a pre-filtered state. To prepare each epoch for the model, each 1.28-second window was normalised using a StandardScaler fitted on the training data only, as given in Eq. (1):

$$x_{norm} = \frac{x - \mu}{\sigma} \tag{1}$$

The same scaler was then applied to the validation and test data of the EEG signals without refitting.

To prevent any information from leaking between the training and test data of the EEG signals we split the data at the file level. The files from each category of the EEG signals were assigned to training, validation and test partitions before any epoch extraction. This means that all the windows from a given MAT file of the EEG signals are in the partition.

A global random seed was fixed for Python, NumPy, PyTorch and CUDA to make sure the results are reproducible. The training data of the EEG signals was augmented with four operations: noise, amplitude scaling, temporal shifting and frequency masking. The augmentation was only applied to the training data of the EEG signals not to the validation or test data of the EEG signals.

Table 1: File-level data partitioning and resulting window counts

Seizure Stage	Train Files	Val Files	Test Files	Windows (T/V/T)
Interictal	32	8	10	416 / 104 / 130
Preictal	32	8	10	416 / 104 / 130
Ictal	32	8	10	416 / 104 / 130
Total	96	24	30	1248 / 312 / 390

2.3. Proposed 1D-ResNet Architecture

The model takes a one single-channel EEG epoch as input. Outputs a probability distribution over the three seizure stage classes. The model has four depthwise blocks, followed by two blocks a global average pooling layer and a classification head. No handcrafted features or frequency-domain transforms are needed for the model of the EEG signals.

2.3.1. Depthwise Convolutional Block

Each block applies a depthwise separable 1D convolution, batch normalization, GELU activation and channel-wise dropout to the EEG signals. Depthwise separable convolution splits a convolution into two steps: a depthwise spatial convolution that processes each input channel c independently (Eq. 2), followed by a pointwise 1×1 convolution that mixes across channels (Eq. 3):

$$D(x) = W_c * x_c \tag{2}$$

$$y = W_p \cdot [D(x)_1, \ldots, D(x)_C] \tag{3}$$

This factorization substantially reduces parameter count while preserving temporal feature extraction, which matters when training data is limited as it is here [20]. GELU activation (Eq. 4) provides smoother gradient flow than ReLU, supporting stable optimisation in this relatively shallow network. Channel-wise dropout drops entire channels rather than individual activations, discouraging co-adaptation between feature channels.

$$GELU(x) = x \cdot \Phi(x) \tag{4}$$

2.3.2. Residual Block

Each residual block applies two depthwise convolutional layers in sequence, then adds the original input via a skip connection (Eq. 5):

$$F(X) = H(X) + X \tag{5}$$

where $H(X)$ is the residual learned by the two convolutional layers and X is the skip-connected input. Residual connections allow gradients to flow freely through layers

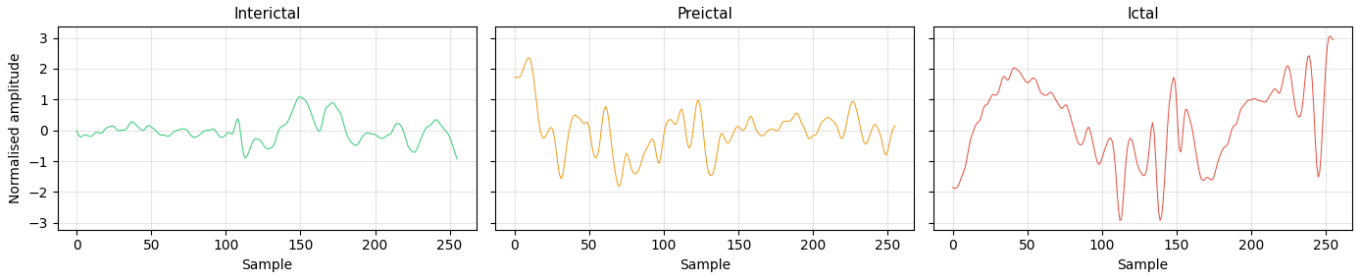


Figure 1: Representative post-normalisation EEG window for each seizure stage (interictal, preictal, ictal).

and prevent information loss as the network deepens [18]. The output $F(X)$ uses the original input X and the output, from the two layers $H(X)$. This helps in making the model learn better from EEG signals. This allows gradients to flow freely through layers and helps the network learn residual corrections rather than full feature mappings of the EEG signals. Practically the skip connections carry level temporal features of the EEG signals, such as waveform morphology, amplitude envelope into deeper layers while the convolutional path refines higher-level stage-discriminative patterns of the EEG signals.

2.3.3. Global Average Pooling and Classification Head

After the residual block global average pooling compresses the temporal dimension by averaging each feature map across the full sequence of the EEG signals. This produces a fixed-length feature vector regardless of input length cuts parameters compared to flattening and provides implicit regularization by averaging over time rather than retaining every localized activation of the EEG signals.

The classification head maps the GAP output to three logits via a linear layer followed by softmax (Eq. 6), producing final class probabilities:

$$p_k = \frac{\exp(z_k)}{\sum_j \exp(z_j)} \tag{6}$$

Softmax output probability for class k over $K = 3$ classes.

Label smoothing (Eq. 7) replaces hard one-hot targets with soft distributions, curbing overconfidence and improving calibration:

$$\tilde{y} = (1 - \varepsilon) \cdot y + \frac{\varepsilon}{K} \tag{7}$$

Label smoothing: $\varepsilon = 0.1$, $K = 3$ classes; y is the hard one-hot target curbing overconfidence and improving probability calibration of the EEG signals. Total model parameters: 981,000.

2.4. Training Configuration

The model was trained using AdamW [19]. It used a learning rate of 1×10^{-3}, betas (0.9, 0.999) and weight decay 5×10^{-4}. The learning rate followed a cosine annealing schedule with warm restarts (Eq. 8), preventing the optimiser from settling in the first local minimum:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left(1 + \cos\left(\frac{\pi t}{T}\right)\right) \tag{8}$$

Cosine annealing LR schedule; $\eta_{max} = 1 \times 10^{-3}$, $\eta_{min} \approx 0$, $T =$ restart period. Early stopping halted training after 15 consecutive epochs without validation loss improvement, restoring the best checkpoint. The batch size was 32.

2.5. Evaluation Metrics

The proposed model was evaluated on the held-out test set using accuracy, per-class precision, recall, F1-score, macro-averaged F1, and one-versus-rest AUC-ROC. Precision-recall curves and mean average precision were also computed. Macro-averaged F1 weights all three classes equally, which is appropriate given the class distribution in the test set.

3. Simulation Setup

3.1. Implementation Details

All simulations were carried out using Python 3.10 and PyTorch 2.0, with a fixed random seed to ensure reproducibility. Each EEG epoch was fed to the model as a raw 256-sample time series without any frequency transformation or manual feature extraction. The four training augmentation operations were applied independently to each window and gated strictly to training data.

3.2. Results

The test set has 390 windows. 130 For each class. This is balanced across all three seizure stages. The model got it right 90.77 percent of the time. It did really well with a macro F1 of 0.907 and a mean AUC-ROC of 0.982. The mean average precision for all classes was 0.9675.

Table 2: Per-class classification performance on the held-out test set

Class	Precision	Recall	F1-Score	AUC-ROC
Interictal	0.8188	0.9385	0.8746	0.9749
Preictal	0.9273	0.7846	0.8500	0.9724
Ictal	0.9924	1.0000	0.9962	1.0000
Macro Avg	0.91	0.91	0.907	0.982

The model was really good at finding seizures. It was almost perfect with an F1 of 0.9962 and an AUC of 1.00. The Ictal phase has an electrophysiological signature. It

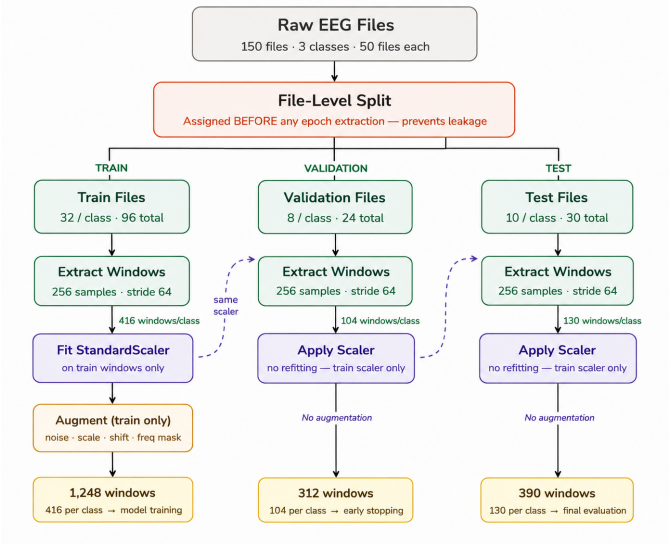


Figure 2: Leakage-free file-level data preprocessing and partitioning pipeline.

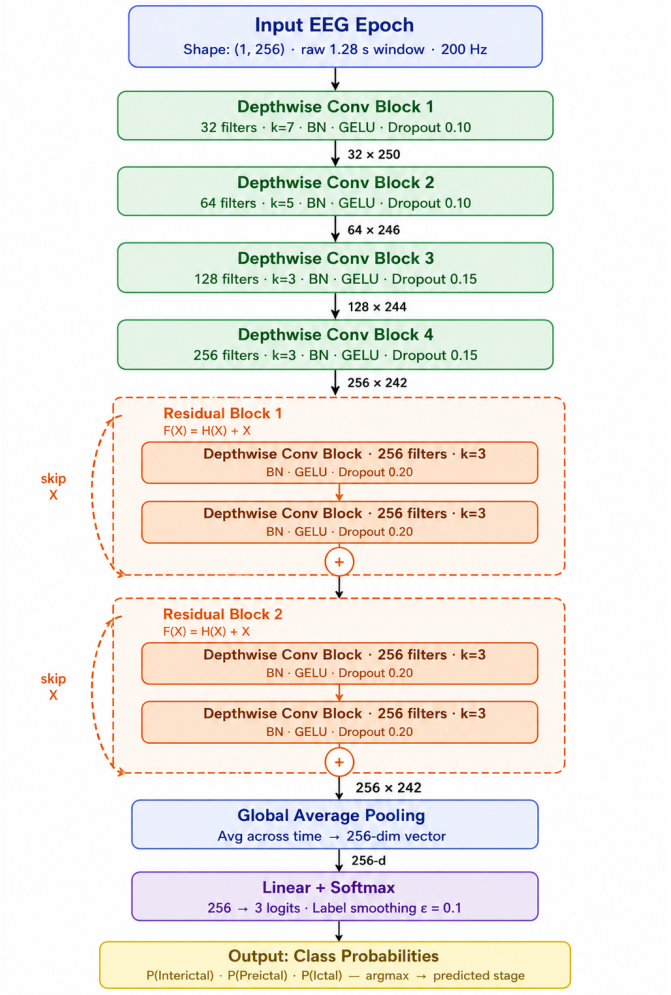


Figure 3: Block diagram of the proposed 1D-ResNet architecture for EEG seizure stage classification.

has high-amplitude and high-frequency oscillatory activity. And the model can find this easily. The model had a time finding Preictal seizures with a recall of 0.7846 because it is really difficult to tell the difference between pre-seizure EEG and the baseline EEG in 1.28-second windows. They can look really similar. The model was not always sure about Interictal seizures with a precision of 0.8188. Some baseline windows were assigned to classes and this makes sense because it can be hard to tell the difference near the boundaries of the stages.

3.3. Comparison with Existing Methods

Table 3 compares the 1D-ResNet against a range of published baselines spanning classical machine learning, convolutional, recurrent, and attention-based architectures, including CNN and CNN-LSTM baselines [4, 14]. Cross-study comparisons require care, since datasets, class definitions, channel counts, and evaluation protocols all differ. Furthermore, recent literature has seen a surge in complex architectures, including Transformer-based models [26,31], attention-guided networks [27], and multi-scale CNNs [29]. Although these advanced architectures achieve high performance, they often do so at the cost of significant computational complexity.

The proposed 1D-ResNet instead aligns with the growing need for lightweight, real-time detection systems [32], demonstrating that a well-regularised, compact residual network can achieve highly competitive results. Crucially, our approach operates under a strictly leakage-free protocol, an evaluation standard that is becoming increasingly critical for robust clinical deployment, as emphasized in recent studies [33], which many prior methods fail to adequately enforce.

Table 3: Comparison with existing seizure stage classification methods

Study	Dataset	Classes	Method	Acc (%)
Massoud et al.	Kaggle iEEG	2	TCNN	60.0
Random Forest + wavelet	CBH-MIT	4	Random Forest	79.7
CNN-GRU + Attention	iEEG private	2	CNN-GRU	82.86
Dutta et al.	Freiburg	4	2D CNN	79.7
Suman et al.	Custom	4	CNN-LSTM	89.0
Proposed Work	Hauz Khas	3	1D-ResNet	90.77

The proposed model outperforms Dutta et al. [4] on Freiburg by 11.07 percentage points in overall accuracy. Notably, Dutta's evaluation used binary one-versus-one comparisons, a simpler task than our simultaneous three-class classification. Against Suman et al. [14], the proposed model achieves slightly higher accuracy (90.77% vs. 89.0%) using a single EEG channel versus nineteen, without the window-level data leakage that likely inflates their numbers.

The remaining baselines in Table 3 use different task definitions, which limits direct comparison but remains informative. Massoud et al. report 60.0% accuracy on a binary two-class task using the Kaggle iEEG dataset with a temporal convolutional network; despite the simpler two-way decision, the proposed model's 90.77% on a harder three-class problem is substantially higher. The Random Forest + wavelet baseline, evaluated on CHB-MIT with

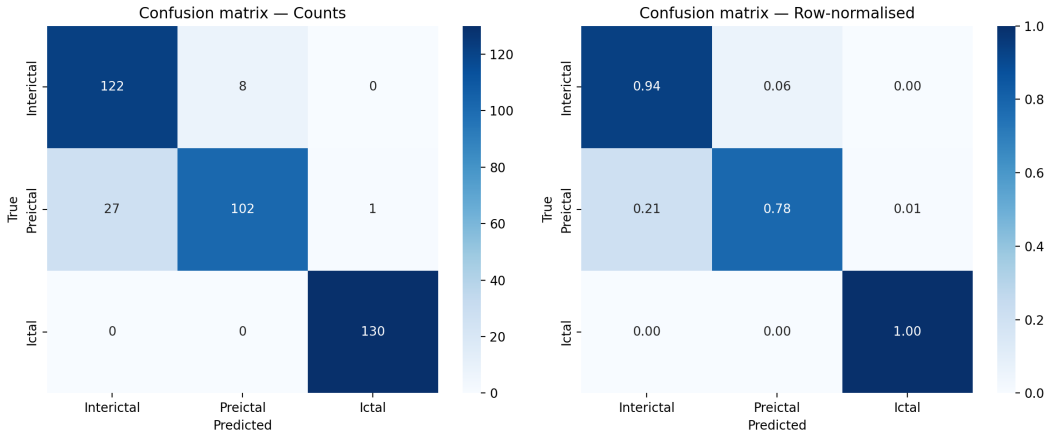


Figure 4: Confusion matrices on the held-out test set: raw counts (left) and row-normalised (right).

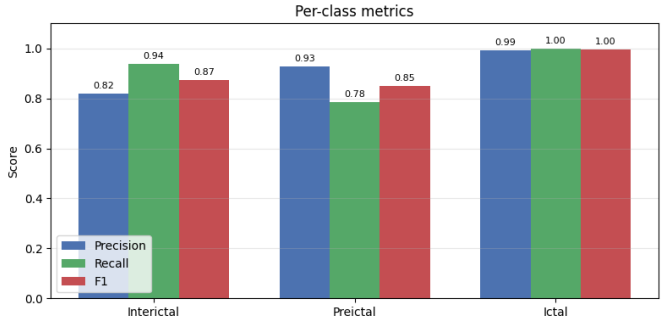


Figure 5: Per-class precision, recall, and F1-score on the held-out test set.

four classes, reaches 79.7% using handcrafted wavelet features rather than learned representations. The proposed model outperforms it by 11.07 percentage points while removing manual feature engineering entirely. CNN-GRU + Attention achieves 82.86% on a private iEEG dataset with only two classes; the proposed model still exceeds this by 7.91 percentage points on the harder three-class task, indicating the depthwise residual architecture remains competitive against attention-based designs.

3.4. Overfitting Analysis

The training and validation accuracy gap was 6.0 percent at the end. This is a little above the target of 5.0 percent. The training stopped early at epoch 78. The validation accuracy was the best, at epoch 63. We tried a thing to make it better. We used channel- dropout and weight decay and label smoothing and training-only augmentation. All of these things helped to reduce the gap. The gap was compared to a baseline that did not have any of these things.

The remaining gap comes down to a fundamental constraint: roughly 981,000 parameters trained on approximately 1,248 training windows (416 per class) is a tight fit. Closing this generalisation gap further, through regularisation strategies or expanded data collection, remains a direction for future work.

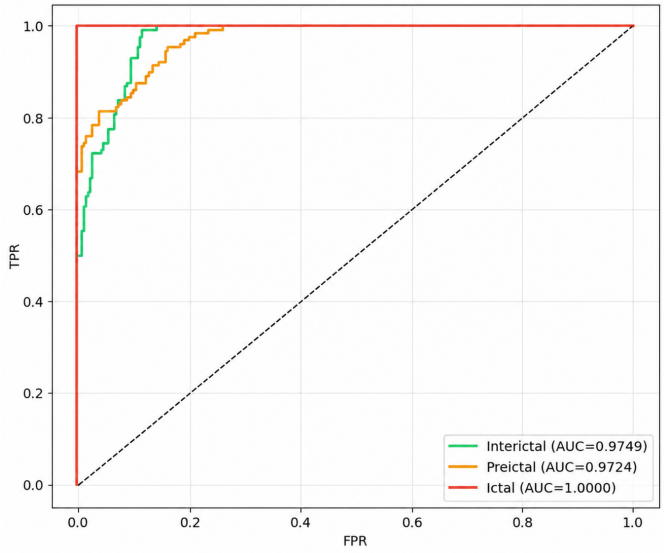


Figure 6: One-vs-rest ROC curves for each seizure stage.

4. Discussion

The near-perfect ictal detection result (F1 = 0.9962, AUC = 1.00) is consistent with what the broader EEG literature consistently finds [4, 14]. The ictal phase is electrophysiologically distinctive: it is readily identifiable once the relevant pattern is known, and local temporal feature detectors in a CNN are well-suited to catching the sharp, high-amplitude waveforms it produces. This result is encouraging but not surprising. The clinically meaningful result is the preictal recall of 0.7846. Roughly one in five preictal windows was missed, mostly misclassified as interictal.

This isn't a problem unique to this model or architecture. The pattern shows up consistently across published methods and datasets [4, 14]: preictal EEG lacks the dramatic signature of active seizure, and the gradual, patient-specific changes that precede seizure onsetunfold over minutes, which is far longer than the 1.28-second window used here can capture. The most direct path toward better preictal recall is extending the model's temporal context, through longer epochs, multi-second input

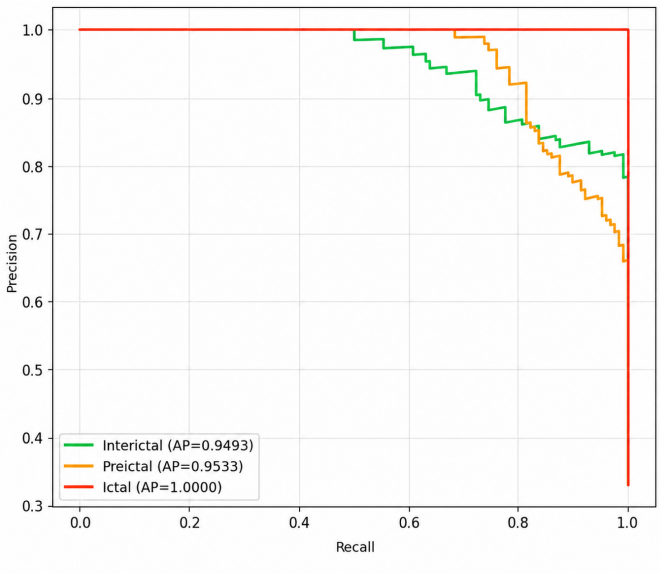


Figure 7: Precision-recall curves for each seizure stage.

sequences, or recurrent architectures that accumulate evidence over time.

Beyond the performance numbers, a key contribution here is the evaluation protocol itself. File-level splitting, train-fitted normalization, training-restricted augmentation, and fixed random seeds together produce trustworthy estimates. Reporting the overfitting gap explicitly, including the fact that it's slightly above our 5.0% target, is transparency that comparable published work rarely offers. In clinical settings, inflated accuracy estimates lead to misplaced confidence in deployment decisions. The 90.77% accuracy and 0.907 macro F1 reported here should be read as conservative, reproducible numbers, not optimistic upper bounds.

Real limitations remain. The dataset covers only ten patients with single-channel EEG. Multi-channel recordings, as used by both comparison studies [4, 14], capture the spatial spread of seizure activity across the scalp that single-channel models simply cannot access. This spatial information is likely particularly useful for preictal detection, where early-stage changes may not be visible in any single electrode.

The absence of the postictal stage also limits clinical scope compared to four-class approaches. And critically, the current file-level split does not guarantee patient-level independence: training and test windows can come from different recordings of the same patient. Patient-wise leave-one-out cross-validation is needed before the model's performance on entirely new patients can be assessed with any confidence, and that's a prerequisite for any realistic clinical deployment.

The broader point this work reinforces: methodological choices around data splitting, normalization, and augmentation scope affect reported performance at least as much as architectural choices. Two models with similar architectures can produce very different accuracy numbers depending on whether leakage-free protocols were followed. The field needs standardized evaluation frameworks, analogous to those that exist in computer vision benchmarking, to enable meaningful cross-study comparisons. The protocol described here offers a reproducible, leakage-free baseline toward that goal. Furthermore, as recent reviews underscore, the integration of explainable artificial intelligence (XAI) is becoming essential to foster clinical trust and utility [28, 34]. Future iterations of this work will aim to incorporate XAI techniques to interpret the model's stage-specific feature extraction visually. Exploring self-supervised representation learning [35] to leverage large volumes of unannotated EEG data also represents a promising avenue to close the generalization gap further.

5. Conclusion

This paper is about a 1D-ResNet for classifying epileptic seizure stages from EEG signals. The researchers tested it on EEG signals from ten patients at the Neurology & Sleep Centre, Hauz Khas, New Delhi. The model was very good at classifying the seizure stages getting it right 90.77 percent of the time. It also did well with measures, like macro F1 and mean AUC-ROC.

The model can look at the raw EEG signals. Figure out what is going on without needing any extra help. It is also good at detecting seizures especially when they are happening. There are things that the researchers think they can work on in the future: They want to try using EEG signals from channels. This might help them detect seizures before they happen by looking at how the signals change across parts of the brain. They want to try out the model on patients. This is to check if it actually works for all kinds of people. They have to be sure that the model does not just work well on the patients they have already tested it on. The model needs to detect seizures in new patients too. They need to see if it works for everyone, not a select few. The goal is to make sure the model is reliable, for all patients. They want to try using types of models that can look at the EEG signals over a longer period of time, and they aim to integrate explainable AI and self-supervised learning methods to enhance both transparency and overall model robustness before they happen.

The main thing the researchers found out is that a simple model can be very good at classifying epileptic seizure stages if it is trained correctly. They think that how you train the model is just as important as what kind of model you use. A trained model can be better, than a more complex model that is not trained as well. The researchers used a residual CNN and trained it on raw EEG signals without using any extra features. This approach worked well and the model was able to match or beat more complex models.

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